MINOR PROJECT -01

TRAFFIC LIGHT CONTROLLER

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DATE OF SUBMISSION:30-08-26

BATCH: VLSI

Introduction

A **Traffic Light Controller** is a digital system that manages the traffic signals at road intersections. It controls the **Red, Yellow, and Green lights** in a proper sequence, helping vehicles and pedestrians move safely and in an organized way.

In this project, the traffic light controller is designed using **Verilog HDL** and works based on a **Finite State Machine (FSM)**. The controller has six different states: **S0, S1, S2, S3, S4, and S5**. Depending on the clock, timer count, and sensor conditions, the controller moves from one state to another and changes the traffic-light signals accordingly.

The project includes both **timer-based and sensor-based** traffic control. The design uses count conditions such as count < 15 and count < 3 to control the timing between different states.

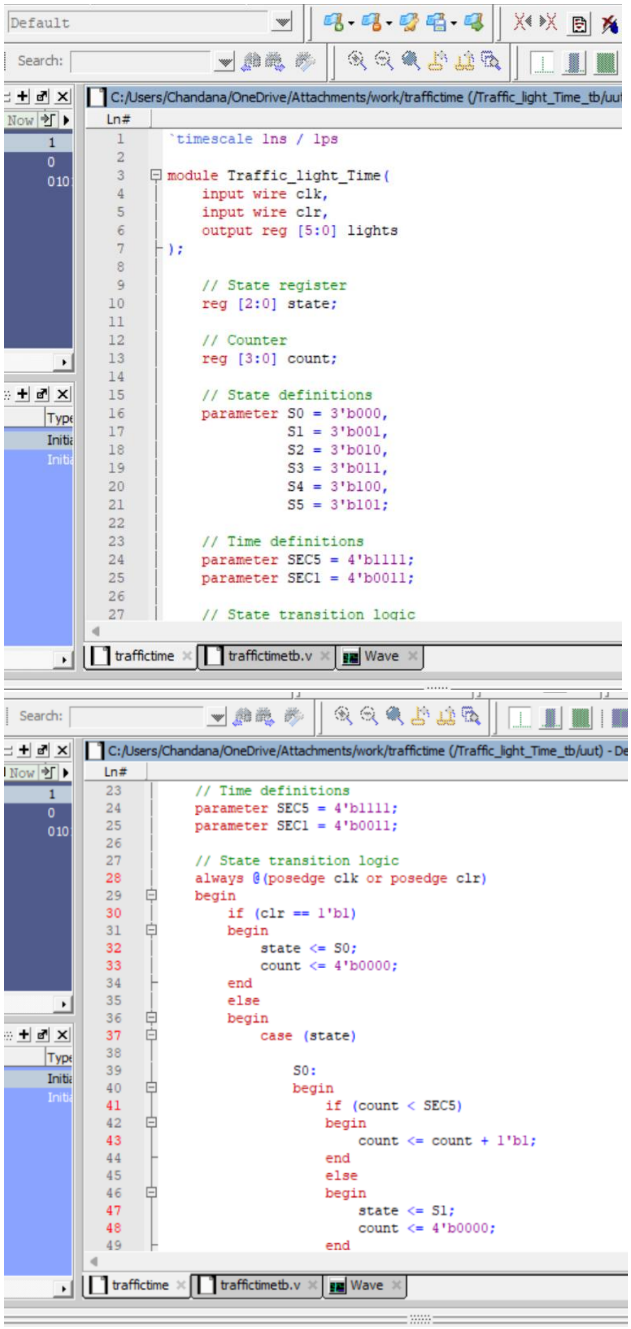
- The project contains 2 codes,

  CODE 1 — Timer-Based Traffic Light Controller

  The first code controls the traffic lights using a **clock signal and a counter**. The counter keeps track of how long the controller has remained in each state. Once the required count is reached, the controller automatically moves to the **next state**, causing the traffic lights to change accordingly.

  CODE 2 — Sensor-Based Traffic Light Controller
  This code controls the traffic lights based on **sensor inputs**. The signals can change depending on the traffic detected by the sensors, making the system more responsive to traffic conditions.

CODE 1 — Timer-Based Traffic Light Controller:



```
`timescale 1ns / 1ps

module Traffic_light_Time(
    input wire clk,
    input wire clr,
    output reg [5:0] lights
);

    // State register
    reg [2:0] state;

    // Counter
    reg [3:0] count;

    // State definitions
    parameter S0 = 3'b000,
              S1 = 3'b001,
              S2 = 3'b010,
              S3 = 3'b011,
              S4 = 3'b100,
              S5 = 3'b101;

    // Time definitions
    parameter SEC5 = 4'b1111;
    parameter SEC1 = 4'b0011;

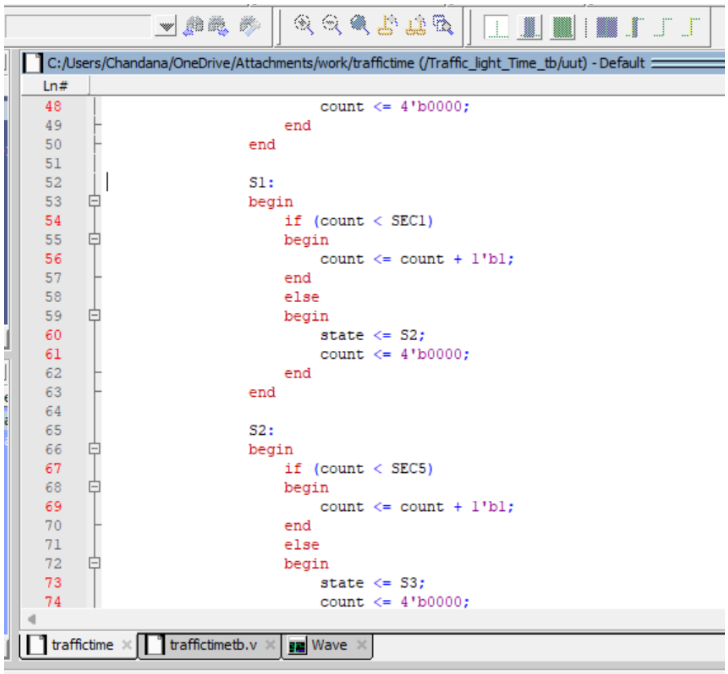
    // State transition logic
```

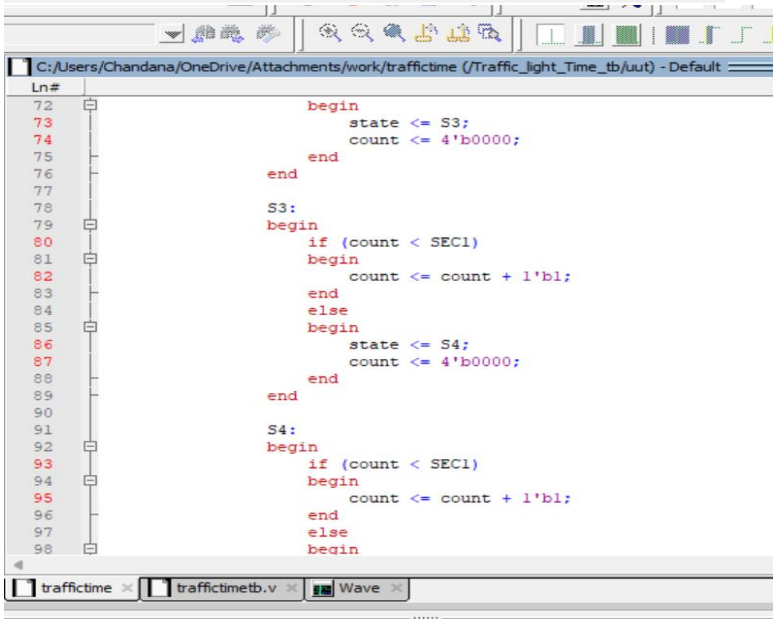


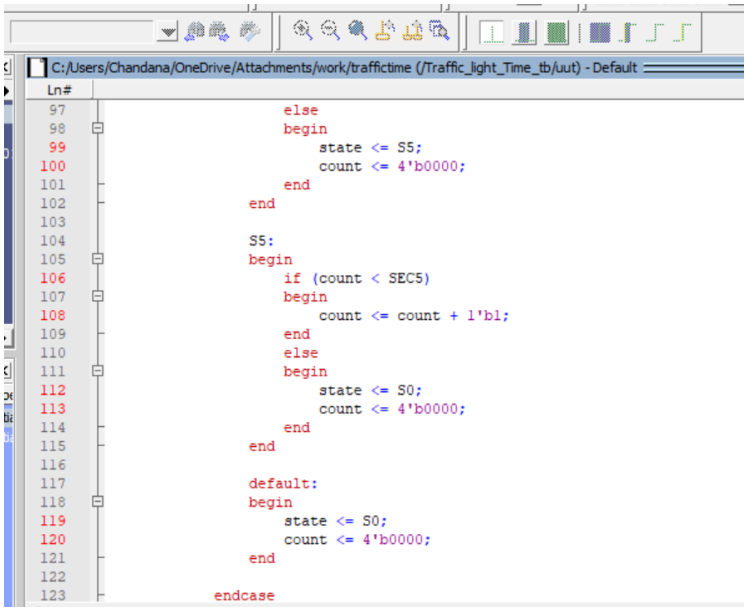
```
    // Time definitions
    parameter SEC5 = 4'b1111;
    parameter SEC1 = 4'b0011;

    // State transition logic
    always @(posedge clk or posedge clr)
    begin
        if (clr == 1'b1)
        begin
            state <= S0;
            count <= 4'b0000;
        end
        else
        begin
            case (state)

                S0:
                begin
                    if (count < SEC5)
                    begin
                        count <= count + 1'b1;
                    end
                    else
                    begin
                        state <= S1;
                        count <= 4'b0000;
                    end
```


C:/Users/Chandana/OneDrive/Attachments/work/traffictime (/Traffic_light_Time_tb/uut) - Default
Ln#
48 count <= 4'b0000;
49 end
50 end
51
52 S1:
53 begin
54 if (count < SEC1)
55 begin
56 count <= count + 1'b1;
57 end
58 else
59 begin
60 state <= S2;
61 count <= 4'b0000;
62 end
63 end
64
65 S2:
66 begin
67 if (count < SEC5)
68 begin
69 count <= count + 1'b1;
70 end
71 else
72 begin
73 state <= S3;
74 count <= 4'b0000;
traffictime
traffictimetb.v
Wave


C:/Users/Chandana/OneDrive/Attachments/work/traffictime (/Traffic_light_Time_tb/uut) - Default
Ln#
72 begin
73 state <= S3;
74 count <= 4'b0000;
75 end
76 end
77
78 S3:
79 begin
80 if (count < SEC1)
81 begin
82 count <= count + 1'b1;
83 end
84 else
85 begin
86 state <= S4;
87 count <= 4'b0000;
88 end
89 end
90
91 S4:
92 begin
93 if (count < SEC1)
94 begin
95 count <= count + 1'b1;
96 end
97 else
98 begin
traffictime
traffictimetb.v
Wave



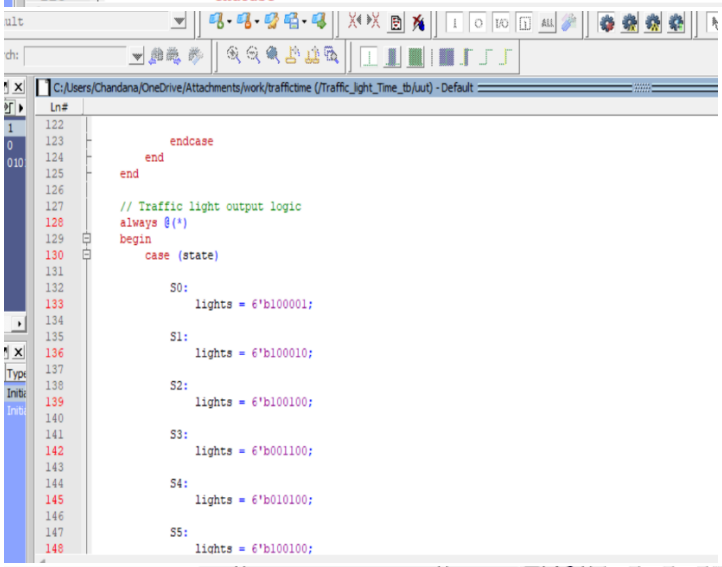
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```
                    else
                    begin
                        state <= S5;
                        count <= 4'b0000;
                    end
                end

                S5:
                begin
                    if (count < SEC5)
                    begin
                        count <= count + 1'b1;
                    end
                    else
                    begin
                        state <= S0;
                        count <= 4'b0000;
                    end
                end

                default:
                begin
                    state <= S0;
                    count <= 4'b0000;
                end

            endcase
```



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```
        endcase
    end
end

// Traffic light output logic
always @(*)
begin
    case (state)

        S0:
            lights = 6'b100001;

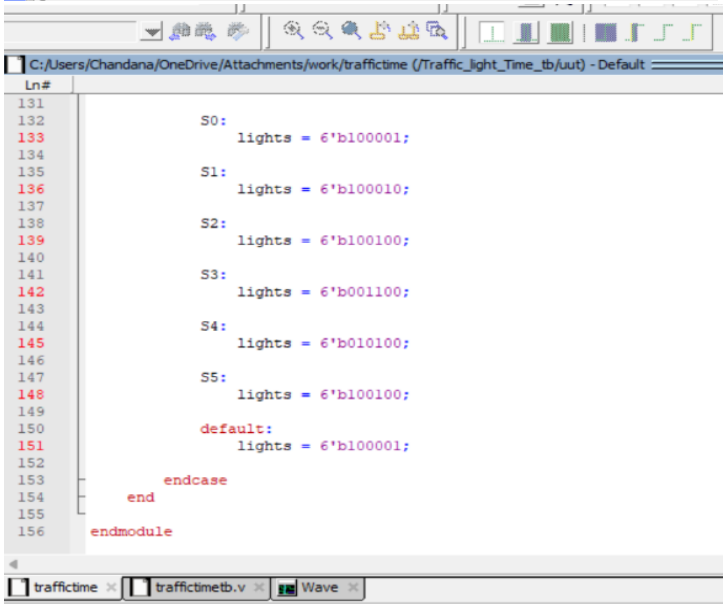
        S1:
            lights = 6'b100010;

        S2:
            lights = 6'b100100;

        S3:
            lights = 6'b001100;

        S4:
            lights = 6'b010100;

        S5:
            lights = 6'b100100;
```



C:/Users/Chandana/OneDrive/Attachments/work/traffictime (/Traffic_light_Time_tb/uut) - Default

```
            S0:
                lights = 6'b100001;

            S1:
                lights = 6'b100010;

            S2:
                lights = 6'b100100;

            S3:
                lights = 6'b001100;

            S4:
                lights = 6'b010100;

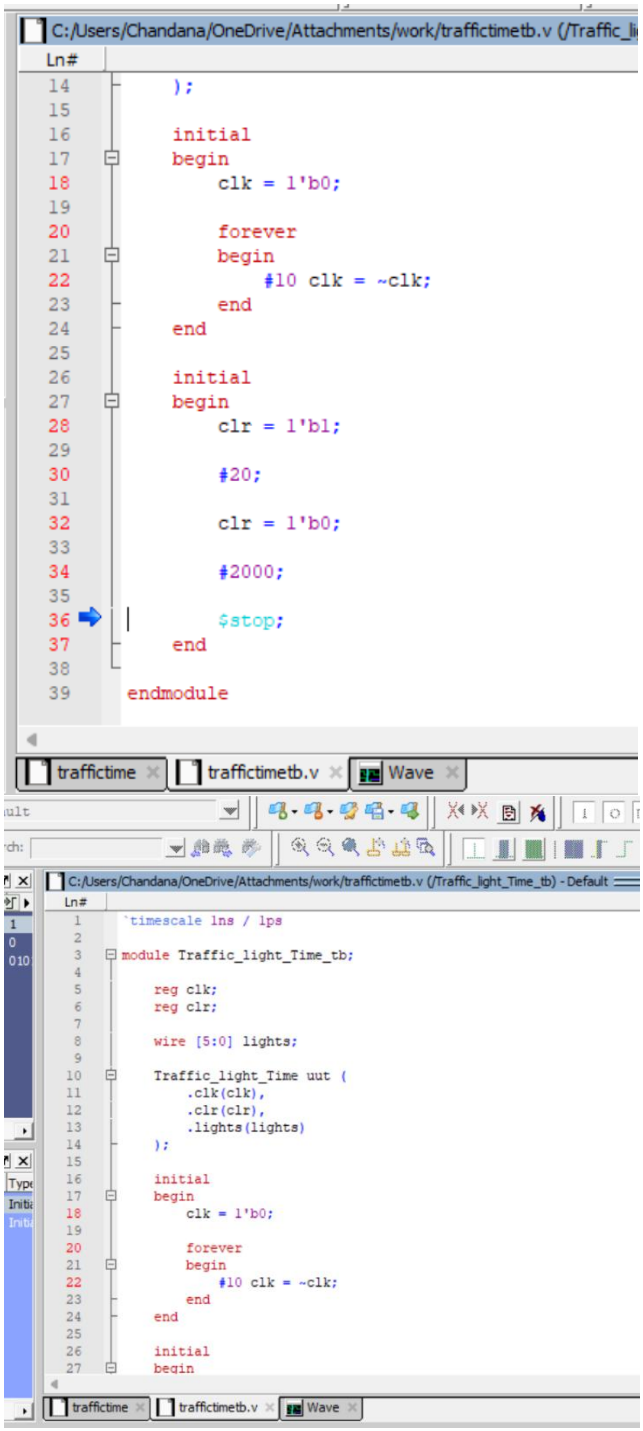
            S5:
                lights = 6'b100100;

            default:
                lights = 6'b100001;

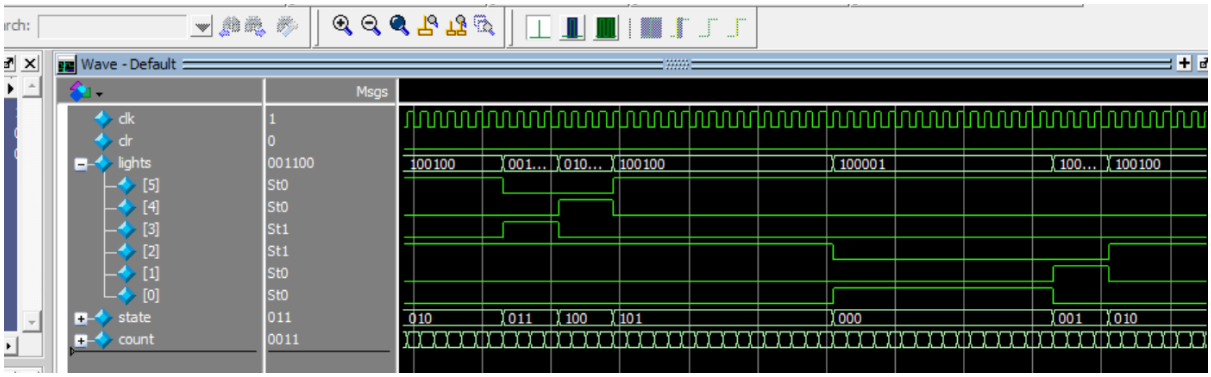
        endcase
    end

endmodule
```

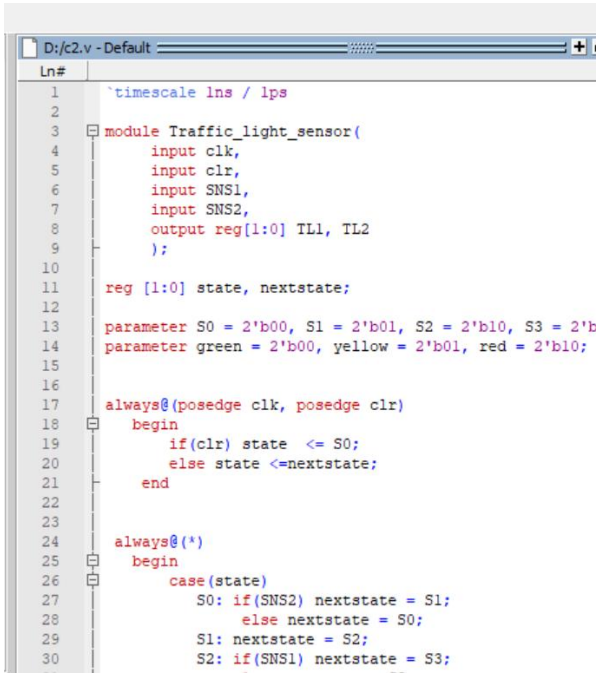
traffictime | traffictimetb.v | Wave

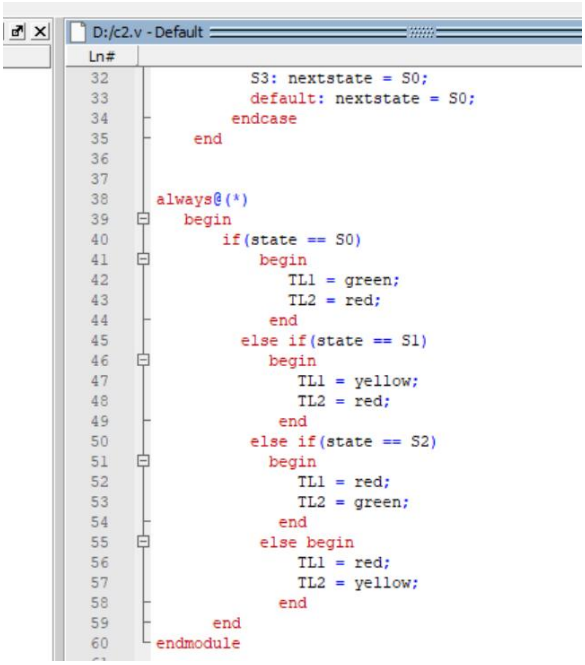

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Ln#
14);
15
16 initial
17 begin
18 clk = 1'b0;
19
20 forever
21 begin
22 #10 clk = ~clk;
23 end
24 end
25
26 initial
27 begin
28 clr = 1'b1;
29
30 #20;
31
32 clr = 1'b0;
33
34 #2000;
35
36 $stop;
37 end
38
39 endmodule
traffictime
traffictimetb.v
Wave
C:/Users/Chandana/OneDrive/Attachments/work/traffictimetb.v (/Traffic_light_Time_tb) - Default
Ln#
1 `timescale 1ns / 1ps
2
3 module Traffic_light_Time_tb;
4
5 reg clk;
6 reg clr;
7
8 wire [5:0] lights;
9
10 Traffic_light_Time uut (
11 .clk(clk),
12 .clr(clr),
13 .lights(lights)
14);
15
16 initial
17 begin
18 clk = 1'b0;
19
20 forever
21 begin
22 #10 clk = ~clk;
23 end
24 end
25
26 initial
27 begin
traffictime
traffictimetb.v
Wave

WAVEFORM 1:



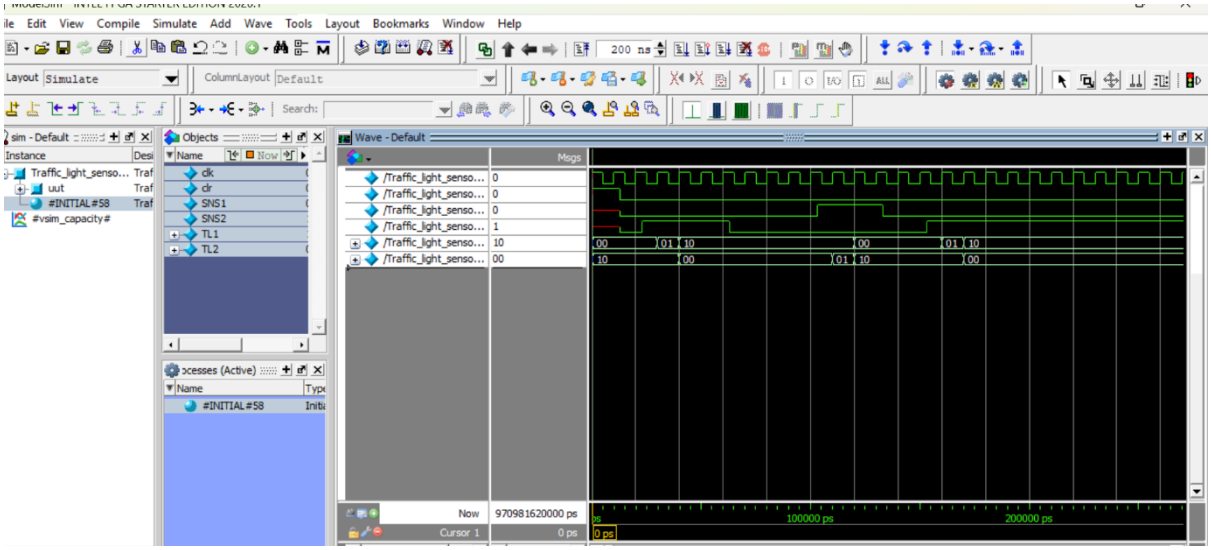
CODE 2 — Sensor-Based Traffic Light Controller:





```
Ln#
 1    `timescale 1ns / 1ps
 2
 3    module Traffic_light_sensor(
 4         input clk,
 5         input clr,
 6         input SNS1,
 7         input SNS2,
 8         output reg[1:0] TL1, TL2
 9         );
10
11    reg [1:0] state, nextstate;
12
13    parameter S0 = 2'b00, S1 = 2'b01, S2 = 2'b1(
14    parameter green = 2'b00, yellow = 2'b01, re(
15
16
17    always@(posedge clk, posedge clr)
18       begin
19           if(clr) state  <= S0;
20           else state <=nextstate;
21        end
22
23
24     always@(*)
25       begin
26           case(state)
27              S0: if(SNS2) nextstate = S1;
28                   else nextstate = S0;
29              S1: nextstate = S2;
30              S2: if(SNS1) nextstate = S3;
22
23
24     always@(*)
25       begin
26           case(state)
27              S0: if(SNS2) nextstate = S1;
28                   else nextstate = S0;
29              S1: nextstate = S2;
30              S2: if(SNS1) nextstate = S3;
31                   else nextstate = S2;
32              S3: nextstate = S0;
33              default: nextstate = S0;
34            endcase
35        end
36
37
38     always@(*)
39       begin
40           if(state == S0)
41               begin
42                  TL1 = green;
43                  TL2 = red;
44               end
45             else if(state == S1)
46                begin
47                   TL1 = yellow;
48                   TL2 = red;
49                end
50              else if(state == S2)
51                begin
52                   TL1 = red;
53                   TL2 = green;
54                end
55              else begin
56                   TL1 = red;
57                   TL2 = yellow;
58                end
59          end
60    endmodule
61
```

WAVEFORM 2:


File Edit View Compile Simulate Add Wave Tools Layout Bookmarks Window Help
Layout Simulate
ColumnLayout Default
Search:
sim - Default
Instance
Traffic_light_senso... Traf
uut Traf
#INITIAL#58 Traf
#vsim_capacity#
Objects
Name
clk
clr
SNS1
SNS2
TL1
TL2
ocesses (Active)
Name
#INITIAL#58
Wave - Default
Msgs
/Traffic_light_senso... 0
/Traffic_light_senso... 0
/Traffic_light_senso... 0
/Traffic_light_senso... 1
/Traffic_light_senso... 10
/Traffic_light_senso... 00
00 01 10 00 01 10
10 00 01 10 00
Now 970981620000 ps
Cursor 1 0 ps
100000 ps
200000 ps
0 ps